**UNIVERSITY OF THE WITWATERSRAND**
**SCHOOL OF BUSINESS SCIENCES**

**INFORMATION SYSTEMS IIA (INFO2001A)**
**TEST 1 — AUGUST 2025_MEMO**

**DURATION:** 1¾ hours (105 minutes) **MARKS:** 55

**INSTRUCTIONS:**
1. This paper consists of **EIGHT (8)** pages. Please ensure that you have them all.
2. You must answer all questions and number them correctly.
3. Sections A and B must be answered in the "**YELLOW**" booklet, while Section B must be answered in the "**GREEN**" booklet. Also put the Section(s) you are answering in the space for "**PAPER No.:**" on each booklet.
4. You may draw diagrams in pencil.
5. You may draw diagrams using landscape page orientation if necessary.
6. You may **NOT** write in the right-hand margins — these are for marking purposes only.

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**Section A**

**1.** D
**2.** A
**3.** D
**4.** A
**5.** C
**6.** C
**7.** B
**8.** C
**9.** D
**10.** A

**Section B**

**1.** ENROLMENT (<u>StudentID</u>, StudentName, StudentEmail, <u>CourseID</u>, CourseName, <u>SemesterID</u>, SemesterStartDate, Grade, InstructorID, InstructorName)

- {<u>StudentID</u>, <u>CourseID</u>, <u>SemesterID</u>} → {StudentName, StudentEmail, CourseName, SemesterStartDate, Grade, InstructorID, InstructorName} (all non-key attributes depend on the full key) (Full/Proper/Functional)✓
- StudentID → StudentName, StudentEmail (partial dependency) ✓
- CourseID → CourseName (partial dependency) ✓
- SemesterID → SemesterStartDate (partial dependency) ✓
- InstructorID → InstructorName (non-key → non-key, giving a transitive path {StudentID, CourseID, SemesterID} → InstructorID → InstructorName) (Transitive) ✓

**2.** STUDENT (<u>StudentID</u> ✓, StudentName, StudentEmail) — Partial Dependency [3NF] ✓

COURSE (<u>CourseID</u> ✓, CourseName) — Partial Dependency [3NF] ✓
SEMESTER (<u>SemesterID</u> ✓, SemesterStartDate) — Partial Dependency [3NF] ✓

| StudentID | CourseID | SemesterID | StudentEmail | Grade | InstructorID | InstructorName |
|---|---|---|---|---|---|---|

3. INSTRUCTOR (<u>InstructorID</u>, StudentName) — Transitive Dependency [3NF]

| StudentID | CourseID | SemesterID | StudentEmail | Grade | InstructorID | InstructorName |
|---|---|---|---|---|---|---|

The following table is in 3NF

| InstructorID | InstructorName |
|---|---|

4. ***Note:** According to the business rules, it was sufficient to resolve a single many-to-many relationship. For this reason, the cardinality between the **SEMESTER** class and the **COURSE** class is defined as **1..2**. However, if you did not assume that an academic year consists of exactly two semesters (and instead modeled the cardinality as **1..***), your solution would still be acceptable, but in that case, you would need to explicitly resolve the relationship.*